

Oscillating Log-Concavity in Geometric Generating Functions

Two Heim–Neuhauser challenges, a general pole criterion,
and the INVERT transform of squares

A research manuscript prepared for Vladimir Reshetnikov

19 September 2026

Abstract

Heim and Neuhauser asked whether the geometric transforms of the square sequence and of each divisor-sum sequence have infinitely many failures of log-concavity. We prove both assertions, in stronger forms. A general meromorphic criterion shows that, after a unique positive dominant pole, a first subdominant circle containing no positive real pole forces both signs of the adjacent Turán determinant to occur with bounded gaps. For divisor weights, a parity identity and elementary radial asymptotics produce a negative pole inside the unit disk. This proves the conclusion for every real divisor exponent and every positive composition weight. For integer power weights, Eulerian-polynomial reciprocity gives a complete qualitative classification: exponents zero and one are eventually log-linear and strictly log-concave, respectively, whereas every exponent at least two exhibits both signs with bounded gaps.

For square weights, the determinant itself satisfies a third-order integer recurrence. We prove that it never vanishes, that every four consecutive indices contain both signs, and that each sign has density one-half within every arithmetic progression. The density proof uses an irrational root angle certified by the Galois group of a cubic of discriminant -107 . The archive includes a dependency-free exact-arithmetic verification program, 100,000 computed signs, and rational certificates for sample Lambert-series inequalities. No finite computation is used to justify an infinite conclusion.

Proven statements and bibliographical status. The targets are precisely Challenges 2 and 3 in Section 3 of [1], arXiv:2302.13327v1, dated 26 February 2023. That version and its challenge page were inspected directly. A targeted search on 19 September 2026 did not locate a prior resolution; this is not an exhaustive certification of priority. The manuscript supplies complete arguments for the stated assertions, but has not undergone independent peer review or formal proof-assistant verification. The known generating function and recurrence of OEIS A033453 are explicitly credited rather than presented as new.

Contents

1	The questions and the resulting theorems	3
2	Geometric transforms and the dominant pole	4
3	A general pole criterion for both signs	5
3.1	A short proof of infinitude using coefficient positivity	5
3.2	A bounded-gap theorem	6
4	Divisor weights: a negative pole for every real exponent	7
4.1	An exact parity identity	8
4.2	Positive radial asymptotics	8
4.3	The negative radial limit and the pole	9
5	Power weights: a complete integer-exponent classification	10
5.1	Even exponents at least two	10
5.2	Odd exponents at least three	11
5.3	The two exceptional low exponents	11
6	Square weights: exact recurrence, sharp gaps, and density	11
6.1	The known sequence and an indexing precaution	11
6.2	An exact determinant recurrence	12
6.3	A single oscillatory term with a decaying positive correction	13
6.4	The sharp four-index block bound	14
6.5	Certifying irrationality of the root angle	14
6.6	Half-density in every arithmetic progression	15
7	Exact computation and reproducibility	16
7.1	Independent checks rather than one implementation twice	16
7.2	Rational certificates for negative Lambert values	16
7.3	Rebuilding the archive contents	17
8	Scope, priority, and remaining questions	17
8.1	What has been established	17
8.2	What is background, and what priority is being claimed	18
8.3	Questions naturally left by the proof	18
A	Proof dependencies and verification boundaries	18

1 The questions and the resulting theorems

For a positive sequence $(a_n)_{n \geq 0}$, write

$$\Delta_n = a_n^2 - a_{n-1}a_{n+1} \quad (n \geq 1). \quad (1.1)$$

Thus $\Delta_n > 0$ means strict log-concavity at n , while $\Delta_n < 0$ means strict log-convexity there. Following the terminology of Heim and Neuhauser, a negative determinant is an *exception* to log-concavity. Their two geometric families are

$$\sum_{n \geq 0} q^{\psi_d(n)} z^n = \frac{1}{1 - \sum_{k \geq 1} k^d z^k}, \quad (1.2)$$

$$\sum_{n \geq 0} q^{\sigma_d(n)} z^n = \frac{1}{1 - \sum_{k \geq 1} \sigma_d(k) z^k}, \quad \sigma_d(k) = \sum_{\ell | k} \ell^d. \quad (1.3)$$

The constant coefficient in both definitions is 1.

Challenge 2 of [1] predicts infinitely many exceptions and non-exceptions for q^{ψ_d} . Challenge 3 predicts infinitely many exceptions for q^{σ_d} , for every integer $d \geq 0$. These are questions with d fixed and n tending to infinity. They differ from the main large- d , fixed- n results in that paper. We do not address the separate exponential/product-generating-function Challenge 1.

It is useful to retain an additional positive parameter u . Set

$$A_{d,u}^{\psi}(z) = \frac{1}{1 - u \sum_{k \geq 1} k^d z^k}, \quad a_{d,u}^{\psi}(n) = [z^n] A_{d,u}^{\psi}(z), \quad (1.4)$$

$$A_{s,u}^{\sigma}(z) = \frac{1}{1 - u \sum_{k \geq 1} \sigma_s(k) z^k}, \quad a_{s,u}^{\sigma}(n) = [z^n] A_{s,u}^{\sigma}(z), \quad (1.5)$$

where d is a nonnegative integer, s may be any real number, and $u > 0$. The use of a positive parameter in this type of geometric transform is already part of the framework in [2]. For integer $s = d \geq 0$ and integer $u > 0$, these are integer sequences with a colored-composition interpretation. Allowing real s or u gives positive real sequences.

Definition 1.1. A set $S \subseteq \mathbb{N}$ has *eventually bounded gaps* if there are integers $N, L \geq 1$ such that each block $\{n, n+1, \dots, n+L-1\}$ with $n \geq N$ contains an element of S .

This is stronger than infinitude, and it implies positive lower natural density. The constants N, L below are allowed to depend on the fixed parameters; no uniformity in d, s, u is asserted.

Theorem 1.2 (Divisor weights). *For every $s \in \mathbb{R}$ and $u > 0$, the sequence $a_{s,u}^{\sigma}(n)$ has strict log-concavity and strict log-convexity at sets of indices that both have eventually bounded gaps. In particular, Challenge 3 of [1] holds for every $d \geq 0$, and infinitely many strict non-exceptions occur as well.*

Theorem 1.3 (Classification for integer power weights). *For $u > 0$, let Δ_n be the determinant of $a_{d,u}^{\psi}(n)$. Then*

d	Δ_1	<i>behavior for $n \geq 2$</i>	(1.6)
0	$-u$	$\Delta_n = 0$	
1	$-2u$	$\Delta_n = u^2$	
$d \geq 2$	$-u 2^d$	<i>both strict signs have eventually bounded gaps.</i>	

Consequently, Challenge 2 holds, and its infinitude conclusion extends to every integer power exponent $d \geq 2$ and every $u > 0$.

The entry $\Delta_1 = -u2^d$ follows directly from $a_0 = 1$, $a_1 = u$, and $a_2 = u^2 + u2^d$.

Theorem 1.4 (Sharp square-weighted conclusions). *Let $a_n = a_{2,1}^\psi(n)$ and define Δ_n by (1.1). Then:*

- (i) $\Delta_1 = -4$, and Δ_n is odd for every $n \geq 2$. In particular, equality never occurs.
- (ii) Every four consecutive positive indices contain both a positive and a negative determinant. Runs of three of either sign do occur, so this four-index block length is optimal.
- (iii) For every modulus $m \geq 1$ and residue class $r \pmod{m}$, each strict sign has relative natural density $1/2$ among the positive indices in that class. Equivalently,

$$\lim_{X \rightarrow \infty} \frac{1}{X} \#\{1 \leq n \leq X : n \equiv r \pmod{m}, \Delta_n > 0\} = \frac{1}{2m}, \quad (1.7)$$

and the same limit holds with $\Delta_n < 0$.

The sign sequence is not eventually periodic.

Theorems 1.2 and 1.3 are proved in Sections 4 and 5, using the general analytic criterion of Section 3. The stronger square-case statements occupy Section 6.

2 Geometric transforms and the dominant pole

Let

$$G(z) = \sum_{k \geq 1} g_k z^k, \quad A(z) = \frac{1}{1 - uG(z)} = \sum_{n \geq 0} a_n z^n, \quad (2.1)$$

where all $g_k > 0$ and $u > 0$. Comparing coefficients in $(1 - uG)A = 1$ gives the exact convolution recurrence

$$a_0 = 1, \quad a_n = u \sum_{k=1}^n g_k a_{n-k} \quad (n \geq 1). \quad (2.2)$$

Alternatively, expanding $A = \sum_{j \geq 0} u^j G^j$ gives

$$a_n = \sum_{\substack{k_1 + \dots + k_j = n \\ j \geq 1, k_i \geq 1}} u^j \prod_{i=1}^j g_{k_i} \quad (n \geq 1). \quad (2.3)$$

For integer weights, a part of size k has g_k possible colors, with an additional u choices per part. Formula (2.3) also makes positivity transparent.

Lemma 2.1 (Unique positive dominant pole). *Suppose G is analytic in $|z| < b$ and $G(x) \rightarrow +\infty$ as $x \uparrow b$, where $0 < b < \infty$. There is a unique $\rho \in (0, b)$ with $uG(\rho) = 1$. The function A is meromorphic in $|z| < b$, and its only pole in $|z| \leq \rho$ is the simple pole at ρ . Its principal part can be written*

$$\frac{C}{1 - z/\rho}, \quad C = \frac{1}{u\rho G'(\rho)} > 0. \quad (2.4)$$

In particular, for some $r > \rho$,

$$a_n = C\rho^{-n} + O(r^{-n}), \quad \rho^n a_n \longrightarrow C, \quad \frac{a_n}{a_{n-1}} \longrightarrow \rho^{-1}. \quad (2.5)$$

Proof. Strict positivity of the coefficients makes $G(x)$ strictly increasing on $(0, b)$, with $G(0) = 0$ and the stated divergent limit. This gives the unique ρ and $G'(\rho) > 0$.

If $|z| < \rho$, then $|uG(z)| \leq uG(|z|) < 1$, so there is no pole. If $|z| = \rho$ and $uG(z) = 1$, equality holds in the triangle inequality for $\sum_{k \geq 1} g_k z^k$. All its nonzero summands must have the same argument. In particular, the terms in z and z^2 have the same argument, so $z/\rho = 1$. Thus the dominant circle has just the positive pole. Expanding $1 - uG(z)$ at $z = \rho$ gives (2.4).

After subtracting (2.4), the pole is removed. Zeros of a nonzero analytic function are isolated; hence the difference is analytic on a disk of radius larger than ρ . Cauchy's coefficient estimate gives (2.5). $\square$

The leading term alone does not decide log-concavity. A geometric sequence has identically zero determinant:

$$(C\rho^{-n})^2 - (C\rho^{-(n-1)})(C\rho^{-(n+1)}) = 0. \quad (2.6)$$

Consequently, an exponentially small correction to a_n can determine the entire sign pattern of Δ_n . The decisive information is often the first *subdominant* pole or group of poles, rather than a sharper estimate of the positive leading constant.

3 A general pole criterion for both signs

The next result is the common mechanism behind the two challenges. We first give a short qualitative argument, then a quantitative-in-form argument proving bounded gaps. Neither requires the subdominant poles to be simple or to form a single conjugate pair.

3.1 A short proof of infinitude using coefficient positivity

We use the following elementary form of Pringsheim's principle.

Lemma 3.1 (Positive coefficients force a positive singularity). *A power series with eventually nonnegative real coefficients and finite, positive radius of convergence R is singular at the positive real point R .*

Proof. Subtracting a polynomial reduces to nonnegative coefficients $F(z) = \sum_{n \geq 0} f_n z^n$. Suppose F were analytic at R , with a continuation to $|z - R| < \delta$. Choose $\varepsilon > 0$ so small that $3\varepsilon < \delta$ and $r = R - \varepsilon > 0$. The Taylor series of the continuation at r converges at $r + 2\varepsilon = R + \varepsilon$. Its derivatives at r equal the derivatives of the original series. Nonnegativity permits interchange of the sums:

$$\begin{aligned} \sum_{k \geq 0} \frac{F^{(k)}(r)}{k!} (2\varepsilon)^k &= \sum_{n \geq 0} f_n \sum_{k=0}^n \binom{n}{k} r^{n-k} (2\varepsilon)^k \\ &= \sum_{n \geq 0} f_n (R + \varepsilon)^n < \infty. \end{aligned}$$

This contradicts the radius of convergence. $\square$

Proposition 3.2 (Qualitative pole obstruction). *Let $A(z) = \sum a_n z^n$ have positive real coefficients and a unique smallest-modulus pole $\rho > 0$, simple, with principal part $C/(1 - z/\rho)$ and $C > 0$. Suppose the remainder*

$$H(z) = A(z) - \frac{C}{1 - z/\rho} \quad (3.1)$$

has finite radius of convergence $R > \rho$ but is analytic at the positive point R . Then Δ_n is strictly positive infinitely often and strictly negative infinitely often.

Proof. The pole expansion gives (2.5). If the determinants were eventually nonnegative, the ratios a_n/a_{n-1} would eventually be decreasing to ρ^{-1} . Therefore $b_n = \rho^n a_n$ would eventually be increasing to C , so $b_n \leq C$ eventually. The coefficients of $-H$ would consequently be eventually nonnegative. Lemma 3.1 would force a singularity at R , a contradiction.

If the determinants were eventually nonpositive, the ratios would be increasing to ρ^{-1} , and b_n would be decreasing to C . Then H , rather than $-H$, would have eventually nonnegative coefficients, giving the same contradiction. These two exclusions prove both strict infinitude statements. $\square$

This proof already resolves the infinitude questions once a suitable secondary pole is found. The next theorem extracts more from the same analytic situation.

3.2 A bounded-gap theorem

Theorem 3.3 (Subdominant poles force both signs with bounded gaps). *Let $A(z) = \sum_{n \geq 0} a_n z^n$ be real at real regular points and meromorphic in $|z| < B$. Suppose:*

- (a) *its unique pole of smallest modulus is a simple pole $\rho > 0$, with principal part $C/(1 - z/\rho)$, where $C > 0$;*
- (b) *there is at least one other pole in $|z| < B$;*
- (c) *if $R > \rho$ is the smallest modulus of these other poles, then A has no pole at the positive real point R .*

Then there are $N, L \geq 1$ such that every block of L indices beginning at $n \geq N$ contains both a strict positive and a strict negative value of $a_n^2 - a_{n-1}a_{n+1}$.

Proof. Step 1: isolate the first subdominant terms. There are finitely many poles on $|z| = R$. Let m be their greatest order, and list the poles of this order as $r_j = R e^{i\theta_j}$, $1 \leq j \leq J$. There is no θ_j congruent to zero modulo 2π , by hypothesis (c). Subtracting all principal parts on and inside this circle leaves a function analytic on a strictly larger disk. The coefficient identity

$$[z^n](1 - z/r)^{-k} = \binom{n+k-1}{k-1} r^{-n}$$

therefore yields

$$a_n = C\rho^{-n} + n^{m-1}R^{-n}(T_n + o(1)), \quad T_n = \sum_{j=1}^J c_j e^{-in\theta_j}. \quad (3.2)$$

Here each c_j is nonzero, and conjugate poles have conjugate constants, so T_n is real. The $o(1)$ consists of lower powers of n and exponentially smaller terms. In particular it remains $o(1)$ after shifting n by either 1 or -1 .

Step 2: compute the first surviving determinant term. Put $a_n = C\rho^{-n} + e_n$. Direct expansion gives

$$\Delta_n = C\rho^{-n}(2e_n - \rho e_{n+1} - \rho^{-1}e_{n-1}) + e_n^2 - e_{n-1}e_{n+1}. \quad (3.3)$$

Substituting (3.2), and using $(n \pm 1)^{m-1}/n^{m-1} = 1 + O(1/n)$, gives

$$\Delta_n = C(\rho R)^{-n} n^{m-1} (U_n + o(1)), \quad (3.4)$$

where

$$U_n = \sum_{j=1}^J v_j e^{-in\theta_j}, \quad v_j = c_j \left(2 - \frac{\rho}{r_j} - \frac{r_j}{\rho} \right). \quad (3.5)$$

The quadratic remainder in (3.3), divided by the positive scale in (3.4), is $O(n^{m-1}(\rho/R)^n)$, which tends to zero. No v_j vanishes: $2 - x - x^{-1} = 0$ implies $x = 1$, whereas $r_j \neq \rho$.

Step 3: a nonzero finite oscillation cannot keep one sign on arbitrarily long blocks. Write $\omega_j = e^{-i\theta_j}$. The frequencies are distinct, $|\omega_j| = 1$, and none equals 1. For every fixed $\omega \neq 1$,

$$\left| \frac{1}{L} \sum_{n=M}^{M+L-1} \omega^n \right| \leq \frac{2}{L|1-\omega|}, \quad (3.6)$$

uniformly in the starting index M . Hence the mean of U_n over such blocks tends to zero uniformly in M . Because U_n is real and the list is closed under conjugation, the constant term in U_n^2 is

$$V = \sum_{j=1}^J |v_j|^2 > 0. \quad (3.7)$$

Every other frequency in the square again has uniformly vanishing block average by (3.6). Thus

$$\frac{1}{L} \sum_{n=M}^{M+L-1} U_n \longrightarrow 0, \quad \frac{1}{L} \sum_{n=M}^{M+L-1} U_n^2 \longrightarrow V, \quad (3.8)$$

uniformly in M . Real poles at $-R$ cause no difficulty: their frequency is -1 , and its square correctly contributes to V .

Let $K = \sum_j |v_j|$, so $|U_n| \leq K$, and choose $\eta = V/(8K) > 0$. Fix L large enough that every length- L block has mean U of absolute value at most η and mean U^2 at least $V/2$. If all values on one block were at most η , their positive parts would have mean at most η . Consequently

$$\text{mean}(U^2) \leq K \text{ mean } |U| = K(2 \text{ mean } U_+ - \text{mean } U) \leq 3K\eta = 3V/8,$$

a contradiction. Applying the same argument to $-U$ shows that every such block contains a value above η and one below $-\eta$.

Finally choose N so that the error in (3.4) has absolute value below $\eta/2$ for $n \geq N$. Its prefactor is positive, so the two values in each block retain their respective signs in Δ_n . $\square$

Remark 3.4 (What the hypotheses do and do not say). The theorem does not require that the nearest subdominant pole be negative real. A nearer complex pole is equally useful. It does require exclusion of a *positive* pole on the first subdominant circle: such a pole would contribute a nonoscillating term to U_n , potentially dominating the sign. The power-weighted case $d = 1$ below is a concrete example of that different behavior.

4 Divisor weights: a negative pole for every real exponent

Put

$$L_s(z) = \sum_{n \geq 1} \sigma_s(n) z^n = \sum_{k \geq 1} \frac{k^s z^k}{1 - z^k}, \quad |z| < 1. \quad (4.1)$$

The equality follows by expanding each geometric denominator and grouping terms with the same product of indices. Absolute convergence is justified by the polynomial bound $\sigma_s(n) \leq n^{1+\max(s,0)}$. Since $\sigma_s(n) \geq 1$, $L_s(x) \geq x/(1-x)$ for $0 < x < 1$. Lemma 2.1 therefore applies to $A_{s,u}^\sigma = 1/(1-uL_s)$ with $b = 1$.

It remains to find a second pole before reaching $|z| = 1$. A negative real pole will suffice. The central observation is that, despite the alternating signs of its Taylor expansion on $(-1, 0)$, $L_s(z)$ tends to *positive* infinity as $z \rightarrow -1$ radially.

4.1 An exact parity identity

Lemma 4.1. *For real s , with $\sigma_s(n/2)$ interpreted as zero when n is odd,*

$$\sigma_s(2n) = (1 + 2^s)\sigma_s(n) - 2^s\sigma_s(n/2). \quad (4.2)$$

Consequently,

$$L_s(-z) = -L_s(z) + (2 + 2^{s+1})L_s(z^2) - 2^{s+1}L_s(z^4). \quad (4.3)$$

Proof. Write $n = 2^a m$ with m odd, and put $q = 2^s$. Then $\sigma_s(n) = (1 + q + \cdots + q^a)\sigma_s(m)$. The identity of finite geometric sums

$$1 + q + \cdots + q^{a+1} = (1 + q)(1 + q + \cdots + q^a) - q(1 + q + \cdots + q^{a-1})$$

proves (4.2), including $a = 0$ when the final sum is empty. This argument also covers $s = 0$, so no division by $2^s - 1$ is needed.

The even-indexed part of $L_s(z)$ is $(1 + 2^s)L_s(z^2) - 2^s L_s(z^4)$ by (4.2). Twice this part minus the whole series is $L_s(-z)$, yielding (4.3). $\square$

4.2 Positive radial asymptotics

Lemma 4.2. *As $t \downarrow 0$,*

$$L_s(e^{-t}) = \begin{cases} \Gamma(s+1)\zeta(s+1)t^{-s-1}(1+o(1)), & s > 0, \\ t^{-1}\log(1/t) + O(t^{-1}), & s = 0, \\ \zeta(1-s)t^{-1}(1+o(1)), & s < 0. \end{cases} \quad (4.4)$$

Proof. For $s > 0$, write

$$t^{s+1}L_s(e^{-t}) = t \sum_{k \geq 1} \frac{(kt)^s}{e^{kt} - 1}.$$

This is a Riemann sum for the integrable function $f_s(x) = x^s/(e^x - 1)$ on $(0, \infty)$. For completeness, near zero one has $f_s(x) \leq x^{s-1}$ and $\sum_{k \leq M} k^{s-1} = O_s(M^s)$; hence the portion $kt \leq \varepsilon$ is $O_s(\varepsilon^s)$ uniformly as $t \downarrow 0$. The far tail is uniformly controlled by exponential decay, and the remaining compact interval is handled by ordinary Riemann sums. Finally, nonnegative-term integration gives

$$\int_0^\infty \frac{x^s}{e^x - 1} dx = \sum_{j \geq 1} \int_0^\infty x^s e^{-jx} dx = \Gamma(s+1)\zeta(s+1).$$

For $s = 0$, take $M = \lfloor 1/t \rfloor$. Uniformly for $0 < x \leq 1$, $(e^x - 1)^{-1} = x^{-1} + O(1)$, so

$$t \sum_{k \leq M} \frac{1}{e^{kt} - 1} = \sum_{k \leq M} \frac{1}{k} + O(tM) = \log(1/t) + O(1).$$

For $k > M$, the bound $(e^{kt} - 1)^{-1} \leq e^{-kt}/(1 - e^{-1})$ gives a contribution $O(1)$ to $tL_0(e^{-t})$. This proves the middle case.

For $s < 0$, use

$$tL_s(e^{-t}) = \sum_{k \geq 1} k^{s-1} \frac{kt}{e^{kt} - 1}.$$

The factor on the right lies in $(0, 1)$ and tends to 1 for each fixed k . Since $\sum k^{s-1} = \zeta(1-s) < \infty$, dominated convergence proves the last case. $\square$

4.3 The negative radial limit and the pole

Proposition 4.3. *For every real s ,*

$$L_s(-e^{-t}) = \begin{cases} 2^{-s-1}\Gamma(s+1)\zeta(s+1)t^{-s-1}(1+o(1)), & s > 0, \\ \frac{1}{2}t^{-1}\log(1/t) + O(t^{-1}), & s = 0, \\ 2^{s-1}\zeta(1-s)t^{-1}(1+o(1)), & s < 0. \end{cases} \quad (4.5)$$

In particular, $L_s(-r) \rightarrow +\infty$ as $r \uparrow 1$.

Proof. Insert $z = e^{-t}$ into (4.3) and apply (4.4) at $t, 2t, 4t$. When $s > 0$, the coefficient of the leading term is

$$-1 + (2 + 2^{s+1})2^{-s-1} - 2^{s+1}4^{-s-1} = 2^{-s-1}.$$

When $s < 0$, it is

$$-1 + \frac{1}{2}(2 + 2^{s+1}) - \frac{1}{4}2^{s+1} = 2^{s-1}.$$

For $s = 0$, the coefficient of $t^{-1}\log(1/t)$ is $-1 + 4/2 - 2/4 = 1/2$; replacing $\log(1/t)$ by $\log(1/(2t))$ or $\log(1/(4t))$ only changes the $O(t^{-1})$ term. All displayed leading constants are strictly positive. $\square$

Proof of Theorem 1.2. Fix $s \in \mathbb{R}$ and $u > 0$. The function $L_s(-r)$ is continuous for $0 \leq r < 1$, starts at zero, and tends to positive infinity. Thus some $\beta \in (0, 1)$ satisfies $uL_s(-\beta) = 1$. Since the numerator of $A_{s,u}^\sigma$ is 1, this zero of its denominator is a genuine pole, of whatever finite order it may have.

Lemma 2.1 supplies the unique dominant pole $\rho > 0$, and also gives $\rho < \beta$. If R is the smallest modulus of a nondominant pole, then $\rho < R \leq \beta < 1$. There is no positive pole at R , because $L_s(x)$ is strictly increasing on $(0, 1)$ and crosses $1/u$ exactly once. Theorem 3.3, with $B = 1$, now proves the result. $\square$

Remark 4.4. We did not prove that $-\beta$ is the nearest nondominant pole, or even that the negative solution is unique. Neither assertion is needed. This avoids a potentially difficult global zero-location problem for Lambert series. Also, the argument does not prove the stronger empirical odd/even pattern reported for small nonnegative integer exponents in [1]; it proves the stated infinitude challenge and the bounded-gap strengthening without relying on that pattern.

5 Power weights: a complete integer-exponent classification

For an integer $d \geq 1$, define the Eulerian polynomial $E_d(z)$ by

$$\sum_{n \geq 1} n^d z^n = \frac{z E_d(z)}{(1-z)^{d+1}}. \quad (5.1)$$

We need only a few elementary properties, which we record with a proof.

Lemma 5.1. *The polynomial E_d has degree $d - 1$, strictly positive coefficients, $E_d(1) = d!$, and*

$$E_d(z) = z^{d-1} E_d(1/z). \quad (5.2)$$

Moreover,

$$E_1(z) = 1, \quad E_{d+1}(z) = (1 + dz)E_d(z) + z(1 - z)E'_d(z). \quad (5.3)$$

Proof. The case $d = 1$ is $\sum n z^n = z/(1 - z)^2$. Applying $z d/dz$ to (5.1) proves (5.3) and the next generating function. If $E_d(z) = \sum_{j=0}^{d-1} e_{d,j} z^j$, its coefficient recurrence is

$$e_{d+1,j} = (j + 1)e_{d,j} + (d - j + 1)e_{d,j-1},$$

with out-of-range entries zero. Starting from $e_{1,0} = 1$, this proves positivity, the degree, and symmetry under $j \mapsto d - j$ for E_{d+1} by induction. The symmetry is (5.2). Putting $z = 1$ in (5.3) gives $E_{d+1}(1) = (d + 1)E_d(1)$, so $E_d(1) = d!$. $\square$

The geometric transform becomes the rational function

$$A_{d,u}^\psi(z) = \frac{(1 - z)^{d+1}}{D_{d,u}(z)}, \quad D_{d,u}(z) = (1 - z)^{d+1} - uz E_d(z). \quad (5.4)$$

The denominator has degree $d + 1$, constant coefficient 1, and leading coefficient $(-1)^{d+1}$. Its value at 1 is $-ud! \neq 0$. The numerator therefore shares no zero with the denominator: every root of $D_{d,u}$ is a genuine pole. In particular, $z = 1$ is a zero of the rational function, not an extra pole.

There is a unique positive pole $\rho \in (0, 1)$, and it is the unique smallest-modulus pole, by Lemma 2.1 applied inside the unit disk. To invoke Theorem 3.3, we inspect possible positive poles outside it.

5.1 Even exponents at least two

For even $d \geq 2$ and real $x > 1$, the numerator $x E_d(x)$ is positive and $(1 - x)^{d+1}$ is negative. Thus the rational continuation of $\sum n^d z^n$ is negative at x , and

$$1 - u \frac{x E_d(x)}{(1 - x)^{d+1}} > 0.$$

There is no positive pole beyond 1. Since $D_{d,u}$ has degree $d + 1 \geq 3$ and the positive pole ρ is simple, other poles exist, but none of them is positive. Theorem 3.3 applies to this rational function in a disk large enough to include its first nondominant circle.

5.2 Odd exponents at least three

For odd d , $d + 1$ is even and (5.2) gives

$$D_{d,u}(z) = z^{d+1}D_{d,u}(1/z). \quad (5.5)$$

The roots are therefore closed under inversion, with multiplicities. The only positive root below 1 is ρ , so the only positive root above 1 is $1/\rho$; there is no root at 1. Remove these two simple roots. When $d \geq 3$, at least two roots remain, and their multiset is still closed under inversion. At least one of them has modulus at most 1; otherwise its reciprocal would violate the same claim. Consequently the nearest nondominant modulus satisfies

$$\rho < R \leq 1 < 1/\rho.$$

The positive root $1/\rho$ cannot lie on that circle, and there are no other positive roots. Again Theorem 3.3 applies.

This argument permits multiple roots and roots on the unit circle. No numerical information about the zeros of Eulerian polynomials is required, and we have not used their stronger real-rootedness property.

5.3 The two exceptional low exponents

For $d = 0$,

$$A_{0,u}^\psi(z) = \frac{1-z}{1-(1+u)z}, \quad a_0 = 1, \quad a_n = u(1+u)^{n-1} \quad (n \geq 1).$$

Thus $\Delta_1 = -u$ and $\Delta_n = 0$ for $n \geq 2$.

For $d = 1$,

$$A_{1,u}^\psi(z) = \frac{(1-z)^2}{1-(u+2)z+z^2} = 1 + \frac{uz}{1-(u+2)z+z^2}. \quad (5.6)$$

The coefficients a_n for $n \geq 1$, extended by the auxiliary value $\tilde{a}_0 = 0$, satisfy

$$\tilde{a}_1 = u, \quad \tilde{a}_n = (u+2)\tilde{a}_{n-1} - \tilde{a}_{n-2} \quad (n \geq 2).$$

For any sequence $x_{n+1} = hx_n - x_{n-1}$, the expression $x_n^2 - x_{n-1}x_{n+1}$ is invariant under shifting n : substitution on both sides gives the same quadratic form. Its value at the auxiliary index 1 here is u^2 . Hence the actual $\Delta_n = u^2$ for $n \geq 2$. At $n = 1$, the actual $a_0 = 1$ instead gives $\Delta_1 = -2u$. Together with the preceding two subsections, this proves Theorem 1.3.

6 Square weights: exact recurrence, sharp gaps, and density

We now fix $d = 2$ and $u = 1$. This is the original Challenge 2 sequence. It is especially suitable for a detailed analysis because its generating function has a cubic denominator.

6.1 The known sequence and an indexing precaution

From $E_2(z) = 1 + z$,

$$A(z) = \frac{(1-z)^3}{1-4z+2z^2-z^3} = 1 + \frac{z(1+z)}{1-4z+2z^2-z^3}. \quad (6.1)$$

The coefficients begin

$$(a_n)_{n \geq 0} = 1, 1, 5, 18, 63, 221, 776, 2725, 9569, 33602, 117995, \dots$$

OEIS A033453 [3] starts at offset 0 with 1, 5, 18, 63, 221, $\dots$. Thus the precise relationship is

$$a_n = A033453(n-1) \quad (n \geq 1), \quad a_0 = 1. \quad (6.2)$$

The OEIS entry already gives the shifted generating function and the recurrence with signature $(4, -2, 1)$, as well as the interpretation by compositions whose parts of size k have k^2 colors. These are established background facts, not claims of novelty here.

In our indexing,

$$a_0 = 1, a_1 = 1, a_2 = 5, a_3 = 18, \quad a_n = 4a_{n-1} - 2a_{n-2} + a_{n-3} \quad (n \geq 4). \quad (6.3)$$

The starting index matters: applying the last recurrence at $n = 3$ with the actual $a_0 = 1$ would incorrectly give 19. The rational series $(z + z^2)/(1 - 4z + 2z^2 - z^3)$ has auxiliary coefficient 0 at $n = 0$ and agrees with a_n for $n \geq 1$. That auxiliary sequence satisfies the homogeneous recurrence from $n = 3$.

6.2 An exact determinant recurrence

Proposition 6.1. *The square-weighted determinant satisfies*

$$\Delta_1 = -4, \quad \Delta_2 = 7, \quad \Delta_3 = 9, \quad \Delta_4 = -9, \quad (6.4)$$

$$\Delta_n = 2\Delta_{n-1} - 4\Delta_{n-2} + \Delta_{n-3} \quad (n \geq 5). \quad (6.5)$$

Equivalently,

$$\sum_{n \geq 2} \Delta_n z^{n-2} = \frac{7 - 5z + z^2}{1 - 2z + 4z^2 - z^3}. \quad (6.6)$$

Every Δ_n with $n \geq 2$ is odd.

Proof. Let $\lambda, \beta, \bar{\beta}$ be the roots of

$$f(x) = x^3 - 4x^2 + 2x - 1. \quad (6.7)$$

The discriminant is -107 , so the roots are distinct, with one real root and one nonreal conjugate pair. Since $f(7/2) = -1/8$ and $f(4) = 7$, the real root satisfies $7/2 < \lambda < 4$. Partial fractions in (6.1) give

$$a_n = \sum_{\xi \in \{\lambda, \beta, \bar{\beta}\}} c_\xi \xi^n, \quad c_\xi = \frac{\xi + 1}{f'(\xi)}, \quad n \geq 1. \quad (6.8)$$

The same formula gives the auxiliary value 0, not the actual 1, at $n = 0$. None of its coefficients is zero.

For a sum $x_n = \sum_i c_i \xi_i^n$, pairwise expansion gives

$$x_n^2 - x_{n-1}x_{n+1} = - \sum_{i < j} c_i c_j (\xi_i - \xi_j)^2 (\xi_i \xi_j)^{n-1}. \quad (6.9)$$

Indeed the diagonal terms cancel, while a pair contributes $c_i c_j (\xi_i \xi_j)^{n-1} (2\xi_i \xi_j - \xi_i^2 - \xi_j^2)$. For the actual a_n , this formula is valid for $n \geq 2$.

Vieta's formulas for (6.7) give

$$\lambda + \beta + \bar{\beta} = 4, \quad \lambda\beta + \lambda\bar{\beta} + \beta\bar{\beta} = 2, \quad \lambda\beta\bar{\beta} = 1.$$

The three pair-products in (6.9) therefore have elementary symmetric functions 2, 4, 1. Their polynomial is $x^3 - 2x^2 + 4x - 1$, proving (6.5) for $n \geq 5$. Direct substitution of the initial a_n gives (6.4); coefficient multiplication then gives (6.6). Finally, reducing (6.5) modulo 2 gives $\Delta_n \equiv \Delta_{n-3} \pmod{2}$. The three starting values $\Delta_2, \Delta_3, \Delta_4$ are odd, proving the parity assertion. $\square$

Some small exact values are useful both as a check and as a warning against guessing eventual alternation.

n	a_n	Δ_n	n	a_n	Δ_n
1	1	-4	7	2725	81
2	5	7	8	9569	311
3	18	9	9	33602	249
4	63	-9	10	117995	-665
5	221	-47	11	414345	-2015
6	776	-49	12	1454992	-1121

In particular, equality is absent at every positive index, not merely eventually.

6.3 A single oscillatory term with a decaying positive correction

Write $\beta = a + ib$ with $b > 0$. The preceding identities yield

$$a = \frac{4 - \lambda}{2} > 0, \quad |\beta|^2 = \beta\bar{\beta} = \lambda^{-1}. \quad (6.10)$$

Put

$$\rho = \lambda^{-1}, \quad \nu = \lambda\beta, \quad |\nu| = \sqrt{\lambda}, \quad (6.11)$$

$$K = -c_\lambda c_\beta (\lambda - \beta)^2, \quad K_0 = 4|c_\beta|^2 b^2 > 0. \quad (6.12)$$

Here $K \neq 0$ and $c_\lambda > 0$. Formula (6.9) becomes the exact expression

$$\Delta_n = 2 \operatorname{Re}(K\nu^{n-1}) + K_0\rho^{n-1} \quad (n \geq 2). \quad (6.13)$$

Let $\theta = \arg \beta = \arg \nu$ and $\phi = \arg K$. Dividing by a positive factor gives

$$\frac{\Delta_n}{2|K|\lambda^{(n-1)/2}} = \cos((n-1)\theta + \phi) + \frac{K_0}{2|K|}\lambda^{-3(n-1)/2}. \quad (6.14)$$

Thus the determinant is governed by a rotation angle, with an exponentially decaying perturbation.

For orientation only, rounded numerical values are

$$\begin{aligned} \lambda &\approx 3.511547141694532, & \rho &\approx 0.284774761564910, \\ \beta &\approx 0.244226429152734 + 0.474476778007327i, \\ \nu &\approx 0.857612619217545 + 1.666147573612060i, \\ \theta &\approx 1.095435949104736, & \theta/\pi &\approx 0.348688092281161. \end{aligned}$$

The subsequent proofs use exact inequalities and algebraic identities, not these decimal approximations.

6.4 The sharp four-index block bound

Lemma 6.2 (Four equally spaced circle points). *Suppose $\pi/3 < \theta < \pi/2$, and put $\delta = -\cos(3\theta/2) > 0$. For every real α , among $\cos(\alpha + j\theta)$, $j = 0, 1, 2, 3$, there is a value at least δ and one at most $-\delta$.*

Proof. The four circle points have circular gaps $\theta, \theta, \theta, 2\pi - 3\theta$. The largest gap is $2\pi - 3\theta$ because $\theta < \pi/2$. Every point of the circle is within angular distance at most $\pi - 3\theta/2$ of one of the four sample points. Apply this to angles 0 and π . Their nearest sample points have cosines at least $\cos(\pi - 3\theta/2) = \delta$ and at most $-\delta$, respectively. $\square$

Proposition 6.3. *Every four consecutive indices $n \geq 1$ contain both determinant signs. The maximum possible length of a run of either sign is exactly three.*

Proof. By (6.10),

$$x := \cos \theta = \frac{(4 - \lambda)\sqrt{\lambda}}{2}. \quad (6.15)$$

The function on the right is decreasing for $\lambda > 4/3$. Since $7/2 < \lambda < 4$, we obtain $0 < x < \sqrt{14}/8 < 1/2$, hence $\pi/3 < \theta < \pi/2$. For $0 < x < 1/2$ the function $4x^3 - 3x$ is decreasing. Therefore

$$\delta^2 = \frac{1 + \cos 3\theta}{2} = \frac{1 + 4x^3 - 3x}{2} > \frac{64 - 17\sqrt{14}}{128} > \frac{1}{400}. \quad (6.16)$$

The last inequality is exact: it is equivalent to $17\sqrt{14} < 1592/25$, which follows on squaring from $4046 < 2534464/625$. Thus $\delta > 1/20$.

We also need a coarse bound on the correction in (6.14). Because $f'(\lambda) = |\lambda - \beta|^2$ and $|f'(\beta)| = 2b|\lambda - \beta|$, the constants simplify to

$$\frac{K_0}{2|K|} = \frac{b|\beta + 1|}{(\lambda + 1)|\lambda - \beta|} < \frac{|\beta|(1 + |\beta|)}{(\lambda + 1)(\lambda - |\beta|)}. \quad (6.17)$$

Here $|\beta| = \lambda^{-1/2} < 3/5$ and $\lambda > 7/2$, so

$$\frac{K_0}{2|K|} < \frac{(3/5)(8/5)}{(9/2)(29/10)} = \frac{32}{435} < \frac{1}{10}.$$

For every $n \geq 2$, the perturbation in (6.14) is consequently strictly less than $1/20$, since $\lambda^{-3/2} < 1/2$.

On any block beginning at $n \geq 2$, Lemma 6.2 gives one cosine at least $\delta > 1/20$ and one at most $-\delta < -1/20$. The perturbations cannot destroy either corresponding sign. The remaining block beginning at $n = 1$ already contains $\Delta_1 = -4$ and $\Delta_2 = 7$. Finally, $\Delta_4, \Delta_5, \Delta_6$ are negative, while $\Delta_7, \Delta_8, \Delta_9$ are positive. Both possible run lengths of three are attained, so the bound is sharp. $\square$

This gives a particularly elementary proof of the infinitude assertion in Challenge 2. The density conclusion requires one further fact: the rotation angle is irrational relative to π .

6.5 Certifying irrationality of the root angle

Lemma 6.4. *For the roots of (6.7), θ/π is irrational.*

Proof. The only possible rational roots of the monic cubic f are 1 and -1 ; neither is a root. Hence f is irreducible over $\mathbb{Q}$. Its discriminant is

$$(-4)^2 2^2 - 4 \cdot 2^3 - 4(-4)^3(-1) - 27(-1)^2 + 18(-4) \cdot 2 \cdot (-1) = -107.$$

The splitting field F therefore has Galois group S_3 : an irreducible cubic has a transitive subgroup of S_3 , and a nonsquare discriminant excludes its subgroup A_3 . The quadratic subfield fixed by A_3 is $\mathbb{Q}(\sqrt{-107})$.

Suppose θ/π were rational. Then $\zeta = \beta/\bar{\beta} = e^{2i\theta}$ would be a root of unity in F . The cyclotomic extension $\mathbb{Q}(\zeta)/\mathbb{Q}$ is Galois and abelian, so the restriction homomorphism from S_3 to its Galois group kills the commutator subgroup A_3 . Thus

$$\mathbb{Q}(\zeta) \subseteq \mathbb{Q}(\sqrt{-107}).$$

A quadratic field can contain a root of unity of order m only if $\varphi(m) \leq 2$, giving $m = 1, 2, 3, 4, 6$. The nonreal possibilities generate $\mathbb{Q}(\sqrt{-3})$ or $\mathbb{Q}(i)$, neither of which equals $\mathbb{Q}(\sqrt{-107})$. Hence $\zeta \in \{1, -1\}$. But $\zeta = 1$ would make β real, while $\zeta = -1$ would make $\beta + \bar{\beta} = 0$, forcing $\lambda = 4$ contrary to $f(4) = 7$. This contradiction proves the lemma. $\square$

6.6 Half-density in every arithmetic progression

Lemma 6.5 (Irrational rotation and a vanishing perturbation). *Suppose $\theta/(2\pi)$ is irrational, $\phi \in \mathbb{R}$, and $\varepsilon_n \rightarrow 0$. Then the sets where $\cos(n\theta + \phi) + \varepsilon_n$ is positive and negative each have natural density $1/2$. The same relative-density conclusion holds after restricting n to any infinite arithmetic progression.*

Proof. For every nonzero integer k ,

$$\frac{1}{N} \sum_{n=1}^N e^{ik(n\theta + \phi)} \rightarrow 0$$

by the finite geometric-sum formula, since $e^{ik\theta} \neq 1$. Approximating continuous periodic functions by trigonometric polynomials, and then approximating interval indicators from above and below, shows that $n\theta + \phi$ is uniformly distributed modulo 2π .

Fix $\eta > 0$. Once $|\varepsilon_n| < \eta$, the perturbed sign agrees with the cosine sign whenever $|\cos(n\theta + \phi)| > \eta$. Uniform distribution gives the density of this exceptional strip as its normalized angular length, which tends to zero as $\eta \downarrow 0$. The positive and negative cosine regions each have length π . Squeezing between the regions with thresholds η and $-\eta$ proves the two densities.

On the progression $n = r + jm$, replace θ by $m\theta$ and the phase by $r\theta + \phi$. Irrationality persists, and the restricted perturbation still tends to zero. The same proof applies. $\square$

Apply Lemma 6.5 to (6.14), using Lemma 6.4. This proves (1.7) and the corresponding negative-density formula. Together with Propositions 6.1 and 6.3, it proves all three parts of Theorem 1.4. If the sign sequence had an eventual period m , it would be eventually constant on each residue class modulo m , contradicting density $1/2$ for both signs within that class.

Notice what has not been needed: no Diophantine lower bound on how closely the angle can approach a zero of the cosine is required. The density argument discards a shrinking-boundary issue using fixed strips and then lets the strip width tend to zero. It does not assert that the cosine and determinant signs agree at every sufficiently large individual index.

7 Exact computation and reproducibility

The program `verify.py` uses only the Python standard library and runs with Python 3.9 or later. It makes no network calls and uses integers or rational numbers for every asserted coefficient, determinant, or certified sign. Its default run was executed with Python 3.13.5. The output is included in `verification-output.txt` and in the `data/` directory.

7.1 Independent checks rather than one implementation twice

The square coefficients are generated in two ways: by the convolution (2.2) with weights k^2 , and by coefficient extraction from the rational function (6.1). They agree through a_{601} . Direct adjacent determinants from those coefficients agree with the independent determinant recurrence (6.5) through Δ_{600} .

The same convolution-versus-rational comparison is performed for $d = 0, \dots, 12$ and $u = 1, 2, 3$, through coefficient a_{121} . Eulerian symmetry, positivity, $E_d(1) = d!$, and the relevant denominator identities are checked through $d = 30$. For divisor weights, a sieve is compared with direct divisor enumeration, and (4.2) is checked for $d = 0, \dots, 10$ and $n = 1, \dots, 300$. These finite checks are useful for catching normalization and indexing errors, especially the exceptional actual value $a_0 = 1$ in the square and linear-power formulas.

The long square computation uses only three live determinant integers, updated by (6.5). This avoids computing a small relative difference of two enormous, nearly equal floating-point products. The integers grow exponentially in magnitude, so constant *number* of live integers does not mean constant bit-space. The algorithm uses $O(N)$ bits of working integer storage up to index N , excluding saved output, and $O(N)$ recurrence steps with increasing operand lengths.

Range $1 \leq n \leq N$	$\Delta_n > 0$	$\Delta_n < 0$	$\Delta_n = 0$	Positive fraction
100	50	50	0	0.500000
1,000	501	499	0	0.501000
10,000	5,003	4,997	0	0.500300
100,000	50,000	50,000	0	0.500000

The exact equality of the last two counts is a property of this finite cutoff, not a claimed formula at every cutoff. The universal density statement is established by the irrational-rotation proof. The longest observed runs have length three, agreeing with the proven sharp four-index block bound. The archive also contains counts in every residue class for moduli 1 through 12.

7.2 Rational certificates for negative Lambert values

For sample nonnegative integer exponents, the program certifies $L_d(-9/10) > 1$ without trusting floating-point evaluation of an alternating series. Let $0 < r < 1$, let

$$S_N = \sum_{n=1}^N \sigma_d(n)(-r)^n, \quad q = r \left(\frac{N+2}{N+1} \right)^{d+1} < 1.$$

Since $\sigma_d(n) \leq n^{d+1}$ and the ratios of successive majorant terms $n^{d+1}r^n$ decrease with n , the remainder obeys

$$|L_d(-r) - S_N| \leq \frac{(N+1)^{d+1}r^{N+1}}{1-q} =: T_N. \quad (7.1)$$

For rational r , both S_N and T_N are exact rational numbers. The assertion $S_N - T_N > 1$ is then a finite, exact certificate of a negative-side crossing of 1.

With $r = 9/10$ and $N = 600$, this succeeds for all $d = 0, \dots, 10$. For example, the following numbers are rounded displays of the exact certificates saved in the JSON file:

d	S_{600} , approximately	T_{600} , approximately
0	7.09052786658	1.93×10^{-24}
1	32.3413228274	1.18×10^{-21}
2	255.356970620	7.18×10^{-19}
5	$1.39444495497 \times 10^6$	1.64×10^{-10}
10	$9.98138686366 \times 10^{13}$	1.40×10^4

Even the coarse final absolute error is negligible relative to that partial sum and still gives a rigorous lower bound greater than 1. These certificates are independent numerical corroboration of the analytic pole mechanism. The proof for arbitrary real s and positive u uses (4.5), not a finite grid of examples.

7.3 Rebuilding the archive contents

From the unpacked directory, run

```
python verify.py
pdflatex -interaction=nonstopmode -halt-on-error article.tex
pdflatex -interaction=nonstopmode -halt-on-error article.tex
```

The first command regenerates the CSV, JSON, and sign data. The two LaTeX runs resolve the table of contents and cross-references. A standard TeX Live installation with the packages listed in the preamble suffices; no external fonts are bundled. A shorter exploratory run is available through `python verify.py -limit 100000`. An optional `numeric_square.py` script uses `mpmath` to reproduce the rounded root and amplitude values; it is not needed for any exact check or proof.

8 Scope, priority, and remaining questions

8.1 What has been established

The original two targets have affirmative proofs. In the divisor family, the argument establishes both strict signs, with bounded gaps, for all real exponents and all positive composition weights. In the power family it classifies every nonnegative integer exponent, again for every positive weight. In the unweighted square case, the exact recurrence, nonvanishing, optimal four-index block bound, and half-density on every arithmetic progression give a substantially more explicit description than infinitude alone.

The general pole theorem is deliberately formulated to survive repeated secondary poles, ties in modulus, and rational dependencies between angles. The square density argument is more specialized: it uses the single conjugate pair and a verified irrational angle. One should not transfer its precise density $1/2$ to every parameter in the larger families merely from the bounded-gap theorem.

8.2 What is background, and what priority is being claimed

The geometric-transform definition, colored-composition interpretation, Eulerian generating functions, partial fractions, Pringsheim's principle, and irrational-rotation equidistribution are standard tools or are explicitly present in the cited sources. They are not presented as new mathematical concepts. The A033453 generating function and recurrence are already recorded in the OEIS. The contribution developed here is the proof of the specified challenges and the accompanying stronger sign statements and classification.

The bibliographical audit was focused rather than exhaustive. The arXiv record inspected lists the 2023 first version; its Section 3 expressly poses the two questions. The publisher metadata identifies a journal publication with DOI 10.1142/S1793042124500192; the revised publisher full text was not available for direct comparison in this session. The challenge numbering in this manuscript therefore refers unambiguously to the inspected arXiv version, not to an uninspected revision. Searches for the paper, its identifier, its authors with geometric log-concavity, and A033453 did not surface a prior proof of these specific assertions. Search coverage was uneven, so this absence is evidence for choosing the problem, not a proof of originality.

A 2026 paper by Alzer and Volkmer [4] was also inspected because its title combines Lambert series and inequalities. Its main log-concavity statement concerns the real-variable function $t \mapsto L_0(1 - e^{-t})$, not the coefficients of $1/(1 - uL_s(z))$. That functional inequality is a different question and does not supply the coefficient-sign conclusions proved here.

8.3 Questions naturally left by the proof

For small nonnegative integer divisor exponents, the computational odd/even pattern is much stronger than mere infinitude. Establishing such an exact parity pattern, where it is valid, would require more specific information about the competing poles or a different direct inequality. The present proof deliberately avoids assuming it.

For the larger parameter families, explicit optimal gap bounds and exact sign densities would require understanding the first subdominant spectrum. Theorem 3.3 provides a route to a computable bound from certified pole locations and residues, but does not compute such constants uniformly in the parameters. Changes in which poles have smallest nondominant modulus can create transitions in the oscillatory pattern.

Finally, higher-order adjacent Hankel or Turán determinants provide a natural extension. Their leading geometric contributions cancel even more extensively. A corresponding analysis would have to identify the first nonvanishing combination of several pole contributions, rather than simply reuse the two-term cross expression (3.3). These are further problems, not premises needed by the results above.

A Proof dependencies and verification boundaries

The main analytic route is

$$\text{positive weights} \implies \text{unique simple positive dominant pole} \implies \text{subdominant analysis.}$$

For divisor weights, the missing nondominant pole is supplied by

$$\text{divisor parity identity} \implies L_s(-r) \rightarrow +\infty \implies \text{a negative pole inside } |z| < 1.$$

For power weights, it is supplied by

$$\text{Eulerian rational function} \implies \text{sign or reciprocal-root analysis} \implies \text{a nonpositive first subdominant circle.}$$

The last phrase means that the circle contains *no pole on the positive real axis*; it does not mean every pole is a nonpositive real number. The common spectral theorem then proves bounded gaps for both signs.

The square-specific route is algebraic:

cubic denominator $\implies$ pair-product determinant recurrence $\implies$ parity and exact cosine formula.

Root bounds and four-point circle geometry prove the sharp gap bound. The Galois group S_3 proves irrationality of the angle, and irrational rotation proves the density theorem.

All numerical roots in the text are illustrative approximations. No root isolation obtained numerically is substituted for the exact inequalities $7/2 < \lambda < 4$, the discriminant calculation, or the Galois argument. The Python tests do not constitute a formalization of complex analysis or Galois theory; they are independent exact checks of the algebraic identities and finite data used to audit the manuscript.

References

- [1] B. Heim and M. Neuhauser, *Log-Concavity of Infinite Product and Infinite Sum Generating Functions*, arXiv:2302.13327v1, 26 February 2023. Challenges 2 and 3 occur in Section 3, printed page 10. <https://arxiv.org/abs/2302.13327>. Journal publication DOI: <https://doi.org/10.1142/S1793042124500192>. The proof targets are cited to the inspected arXiv version.
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- [3] The OEIS Foundation Inc., *The On-Line Encyclopedia of Integer Sequences*, entry A033453, “INVERT transform of squares.” <https://oeis.org/A033453>. Accessed 19 September 2026. The entry’s offset is 0 and its first term corresponds to a_1 in this manuscript.
- [4] H. Alzer and H. W. Volkmer, *Inequalities for the Lambert series*, The Ramanujan Journal **70**, article 1 (2026), published 10 April 2026. <https://doi.org/10.1007/s11139-026-01365-x>. Cited to distinguish functional log-concavity from the coefficient problem studied here.